

Sophie Loupias

engineering student

I am an aspiring engineer in the field of electronics and electrical energy looking to apply my technical skills to solving real problems. Hard-working, adaptable, with the ability to connect with people from different backgrounds, I am a great asset to any team.

CONTACT

sophie.loupias@etu.inp-n7.fr

07 83 58 12 06

31 000 Toulouse

LinkedIn

Languages

English (native)

French (native)

Spanish B1

Korean A1

Software skills

Programming

C ◆◆◆◆

Python ◆◆◆◆

SQL ◆◆◆◇

Assembly language ◆◆◆◇

Web/OS

HTML & CSS ◆◆◆◇

Linux ◆◆◇◇

Simulation & CAD

Excel ◆◆◆◆

MATLAB & Simulink ◆◆◆◇

Civic engagement

Les restos du coeur

- Food distribution and sorting
- Administrative work

Interests

Art: oil and acrylic painting

Music: playing clarinet in a band

Sports: gymnastics, volleyball, tennis, rock climbing

Education

ENSEEIHT: Master's in electronics and electrical energy *2025-2028*

Lycee Bellevue: Preparatory class in physics and math *2023-2025*

Lycee Laperouse: Baccalauréat with highest distinctions *2020-2023*

Canyon Crest Academy: High school (San Diego, CA) *2018-2019*

Internships

Cornerstone Insurance -Virginia, USA *june-july 2026*

- Policy administration: Audited portfolios to identify expiring policies and researched client accounts to maintain accurate database records.
- Client communication: Answered incoming calls and managed email communications, ensuring prompt responses
- Accounts receivable: Conducted email follow-ups on past-due accounts to resolve late payments and ensure continuity of coverage.

Projects

Solar panel MPPT *2024-2025*

- Characterised the photovoltaic modules, generating and analyzing V(I) and P(I) performance curves.
- Modelled and simulated the complete power system circuit and Buck converter in MATLAB/Simulink to validate the algorithm design.
- Programmed and implemented a Maximum Power Point Tracking (MPPT) algorithm on an Arduino microcontroller.

Robot Implementation and Control *2026*

- Designed and implemented memory address decoding and integrated a Nios II processor on an Altera Cyclone V FPGA using Quartus Prime.
- Developed C drivers and modular software for peripherals including 7-segment displays, LEDs, switches, a joystick controller, and DC motors.
- Programmed and validated autonomous and manual control algorithms: polling mode, obstacle avoidance via sensors, and line-following strategies.

Professional experience

Cook & Vendor | farmer's market -France *weekly, 2026-currently*

- Prepared and served food while ensuring good customer service.
- Maintained smooth workflow during peak traffic hours.

Teacher's assistant | elementary school -France *2019*

- Supervised groups of children (ages 3–6), maintaining a safe and structured environment.
- Worked with the educational team to plan creative and developmental activities.